4 **D** For a (perfectly) elastic collision between two bodies, the relative speed of approach is equal to the relative speed of separation.

relative speed of P approaching Q	Option	relative speed of Q separating from P	equal ?

<table>
<tr><td rowspan="4">$(1.30)-(-0.50)$
$=1.80\text{ m s}^{-1}$</td><td>**A**</td><td>$(0.58)-(0.22)=0.36\text{ m s}^{-1}$</td><td>✗</td></tr>
<tr><td>**B**</td><td>$(0.40)-(0.40)=0.00\text{ m s}^{-1}$</td><td>✗</td></tr>
<tr><td>**C**</td><td>$(0.99)-(-0.19)=1.18\text{ m s}^{-1}$</td><td>✗</td></tr>
<tr><td>**D**</td><td>$(1.30)-(-0.50)=1.80\text{ m s}^{-1}$</td><td>✓</td></tr>
</table>

5 **B** By Newton's Third Law of Motion